

WEEK-9 DAY-45 Hands On Keerthivasan R

Problem Statement

A development team is working on a web application project. Multiple developers are contributing code simultaneously. However:

- There is no structured method to manage file changes.
- Developers overwrite each other's work.
- Deleted files cannot be recovered easily.
- There is no proper tracking of who made changes and when.
- Rollback to previous versions is difficult.

The organization requires a structured version control system to manage **CRUD operations (Create, Read, Update, Delete)** on project files using Git.

☒ Proposed Solution

Implement Git to perform and manage CRUD operations effectively.

◇ CRUD Operations in Git Context

1 Create (C)

- Add new files to repository.

Commands:

```
git add filename
git commit -m "Added new file"
```

-

2 Read (R)

- View file history and changes.

Commands:

```
git status
git log
git diff
```

-

3 Update (U)

- Modify existing files and track updates.

Commands:

```
git add filename  
git commit -m "Updated file"
```

-

Delete (D)

- Remove files from repository.

Commands:

```
git rm filename  
git commit -m "Deleted file"
```

-
-

Functional Requirements

- Initialize repository.
 - Track file changes.
 - Commit updates with proper messages.
 - Delete files safely.
 - View history of modifications.
 - Restore previous versions if needed.
-

Non-Functional Requirements

- Maintain version history.
 - Ensure collaboration among team members.
 - Prevent data loss.
 - Support branching and merging.
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Expected Outcome

After implementing CRUD operations in Git:

- Code changes will be properly tracked.
- Collaboration will improve.
- File history can be viewed anytime.
- Deleted files can be restored.
- Project versioning becomes efficient and reliable.

OUTPUT:

```
keerthi@keerthivasan MINGW64 ~
$ cd ~/Desktop

keerthi@keerthivasan MINGW64 ~/Desktop
$ mkdir WebProject

keerthi@keerthivasan MINGW64 ~/Desktop
$ cd WebProject

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject
$ git init
Initialized empty Git repository in C:/Users/keerthi/Desktop/WebProject/.git/

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ echo "<h1>Welcome to Our Web Project</h1>" > index.htm

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ echo "<h1>Welcome to Our Web Project</h1>" > index.html

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ echo "body { font-family: Arial; }" > style.css

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    index.htm
    index.html
    style.css

nothing added to commit but untracked files present (use "git add" to track)

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ git add index.html
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git
touches it

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ git add style.css
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git
touches it

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ git commit -m "Added new project files - index.html and style.css"
[main (root-commit) 33343dd] Added new project files - index.html and style.css
2 files changed, 2 insertions(+)
create mode 100644 index.html
create mode 100644 style.css

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git log --oneline
33343dd (HEAD -> main) Added new project files - index.html and style.css

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    index.htm

nothing added to commit but untracked files present (use "git add" to track)

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ git log
commit 33343dde16acece73c6c2e13642c02ef0afa3418 (HEAD -> main)
Author: keerthivjr <keerthivasankeerthi43@gmail.com>
Date:   Sun Apr 19 22:11:41 2026 +0530

    Added new project files - index.html and style.css

keerthi@keerthivasan MINGW64 ~/Desktop/webProject (main)
$ echo "h1 { color: blue; }" >> style.css

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git diff
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git
touches it
```

```

diff --git a/style.css b/style.css
index c3443ba..7369593 100644
--- a/style.css
+++ b/style.css
@@ -1,2 @@
 body { font-family: Arial; }
+h1 { color: blue; }

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git log --pretty=format:"%h - %an, %ar : %s"
33343dd - keerthivjr, 5 minutes ago : Added new project files - index.html and style.css

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ echo "h1 { color: blue; }" >> style.css

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ echo "<p>This is a paragraph.</p>" >> index.html

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git diff
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git
touches it
diff --git a/index.html b/index.html
index 14513d9..0a4688b 100644
--- a/index.html
+++ b/index.html
@@ -1,2 @@
 <h1>welcome to Our Web Project</h1>
+<p>This is a paragraph.</p>
diff --git a/style.css b/style.css
index c3443ba..ee69572 100644
--- a/style.css
+++ b/style.css
@@ -1,3 @@
 body { font-family: Arial; }
+h1 { color: blue; }
+h1 { color: blue; }

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git add style.css index.html
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git
touches it

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git commit -m "Updated style.css with blue heading color and added paragraph to index.html"
[main a67bfc3] Updated style.css with blue heading color and added paragraph to index.html
2 files changed, 3 insertions(+)

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git log --oneline
a67bfc3 (HEAD -> main) Updated style.css with blue heading color and added paragraph to
index.html
33343dd Added new project files - index.html and style.css

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ echo "This is temporary" > temp.txt

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git add temp.txt
warning: in the working copy of 'temp.txt', LF will be replaced by CRLF the next time Git
touches it

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git commit -m "Added temporary file"
[main 70991e9] Added temporary file
1 file changed, 1 insertion(+)
create mode 100644 temp.txt

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git rm temp.txt
rm 'temp.txt'

keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    deleted:    temp.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)

```

index.htm

```
keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git commit -m "Deleted temporary file temp.txt"
[main e021504] Deleted temporary file temp.txt
1 file changed, 1 deletion(-)
delete mode 100644 temp.txt
```

```
keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ ls -la
total 11
drwxr-xr-x 1 keerthi 197121 0 Apr 19 22:17 ./
drwxr-xr-x 1 keerthi 197121 0 Apr 19 22:09 ../
drwxr-xr-x 1 keerthi 197121 0 Apr 19 22:18 .git/
-rw-r--r-- 1 keerthi 197121 36 Apr 19 22:10 index.htm
-rw-r--r-- 1 keerthi 197121 64 Apr 19 22:16 index.html
-rw-r--r-- 1 keerthi 197121 69 Apr 19 22:16 style.css
```

```
keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$ git log --oneline
e021504 (HEAD -> main) Deleted temporary file temp.txt
70991e9 Added temporary file
a67bfc3 Updated style.css with blue heading color and added paragraph to index.html
33343dd Added new project files - index.html and style.css
```

```
keerthi@keerthivasan MINGW64 ~/Desktop/WebProject (main)
$
```